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dspexp1amlx.mlx | exp1g.m | untitled2

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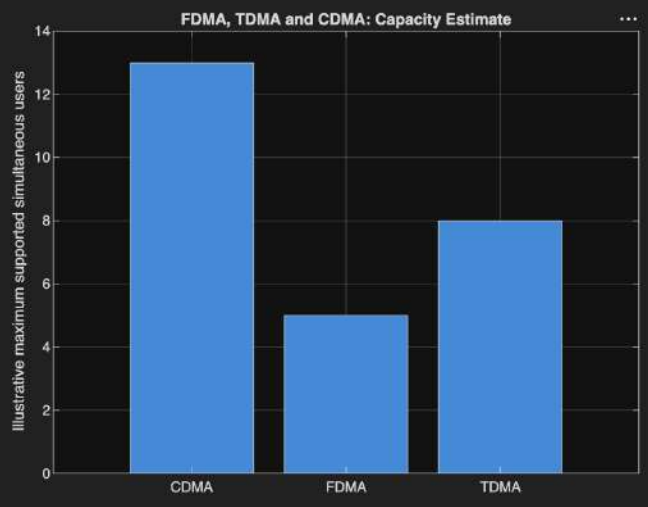
```
fdmaChannelHz = 200e3;
tdmaSlots = 8;
processingGain = 64;
targetSIRdB = 7; %
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB / 10);
cdmaUsers = floor(1 + processingGain / targetSIR);
fprintf('Illustrative maximum supported users\n');
fprintf('FDMA: %d users\n', fdmaUsers);
fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
    cdmaUsers, targetSIRdB);

K = 2:30;
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);
method = ["FDMA"; "TDMA"; "CDMA"];
supportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];

results = table(method, supportedUsers);
disp(results);
figure;
bar(categorical(method), supportedUsers);
ylabel('Illustrative maximum supported simultaneous users');
title('FDMA, TDMA and CDMA: Capacity Estimate');
grid on;
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of simultaneous CDMA users (K)');
ylabel('Estimated SIR (dB)');
title('Estimated CDMA SIR versus Number of Users');
grid on;
hold off;
```

Figures

Figure 1 | Figure 2



Command Window

```
FDMA: 5 users
TDMA: 8 users
CDMA: 13 users at a target SIR of 7.0 dB

    method    supportedUsers
    ______    ______________

    "FDMA"           5
    "TDMA"           8
    "CDMA"          13

>>
```

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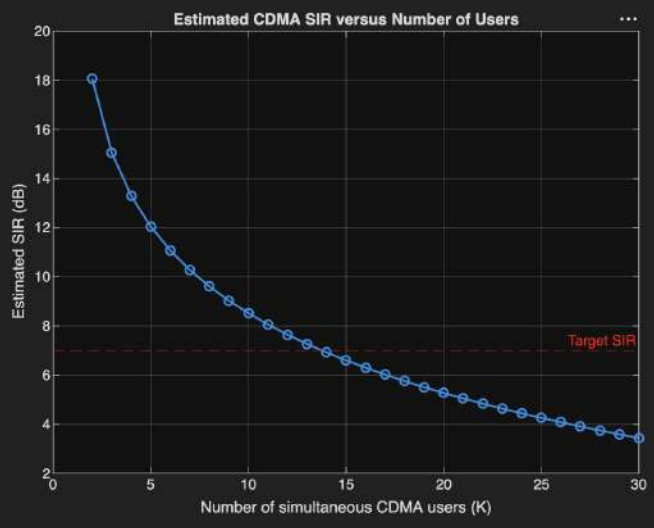
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grid on;
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of simultaneous CDMA users (K)');
ylabel('Estimated SIR (dB)');
title('Estimated CDMA SIR versus Number of Users');
grid on;
hold off;
```

Figures

Figure 1 | Figure 2



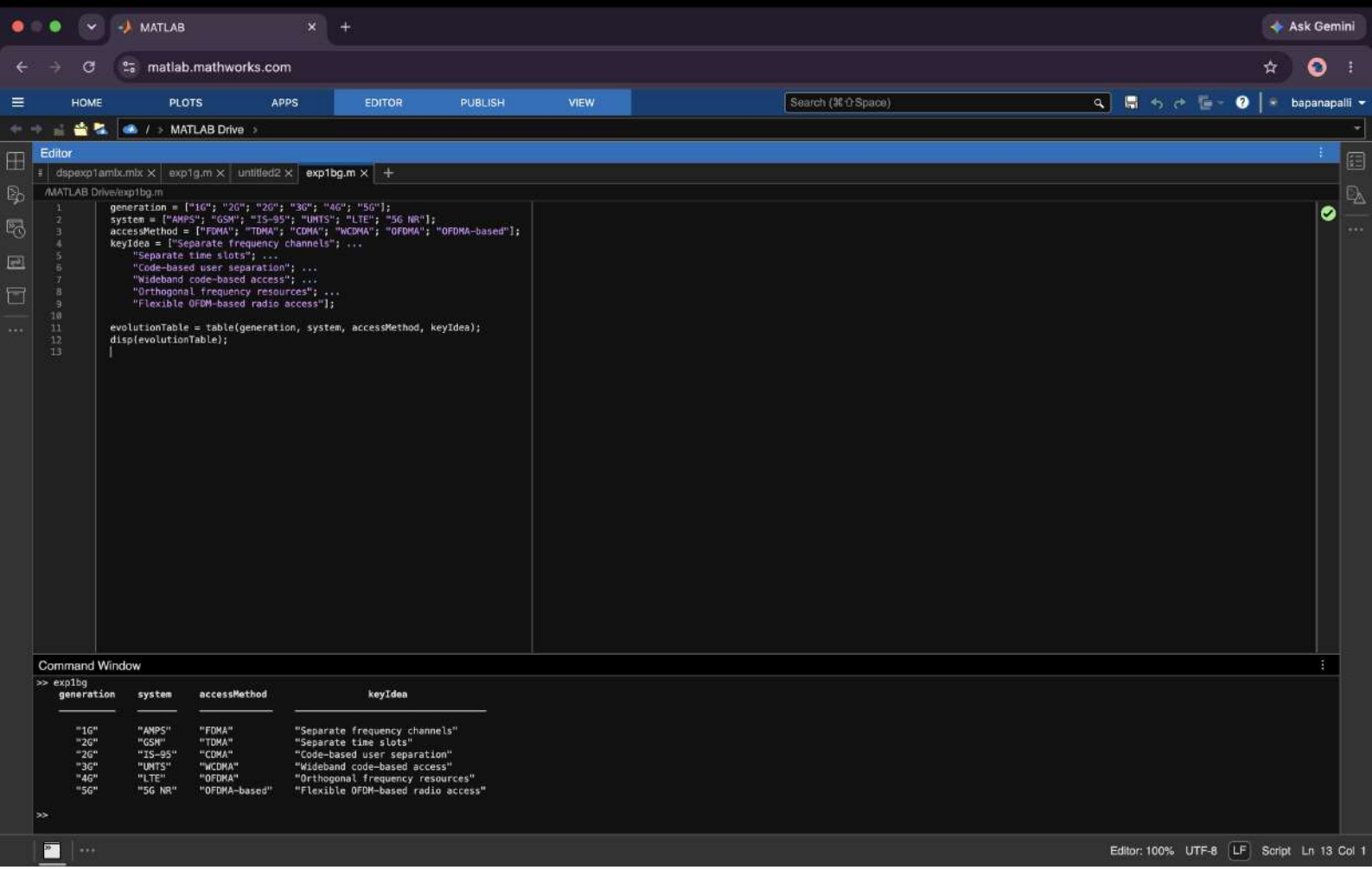
Command Window

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FDMA: 5 users
TDMA: 8 users
CDMA: 13 users at a target SIR of 7.0 dB

    method    supportedUsers
    ______    ______________

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    "TDMA"           8
    "CDMA"          13

>>
```


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exp1bg.m
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generation = ["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];
system = ["AMPS"; "GSM"; "IS-95"; "UMTS"; "LTE"; "5G NR"];
accessMethod = ["FDMA"; "TDMA"; "CDMA"; "WCDMA"; "OFDMA"; "OFDMA-based"];
keyIdea = ["Separate frequency channels"; ...
"Separate time slots"; ...
"Code-based user separation"; ...
"Wideband code-based access"; ...
"Orthogonal frequency resources"; ...
"Flexible OFDM-based radio access"];
evolutionTable = table(generation, system, accessMethod, keyIdea);
disp(evolutionTable);
Command Window
>> exp1bg
generation system accessMethod keyIdea
"1G" "AMPS" "FDMA" "Separate frequency channels"
"2G" "GSM" "TDMA" "Separate time slots"
"2G" "IS-95" "CDMA" "Code-based user separation"
"3G" "UMTS" "WCDMA" "Wideband code-based access"
"4G" "LTE" "OFDMA" "Orthogonal frequency resources"
"5G" "5G NR" "OFDMA-based" "Flexible OFDM-based radio access"
>>
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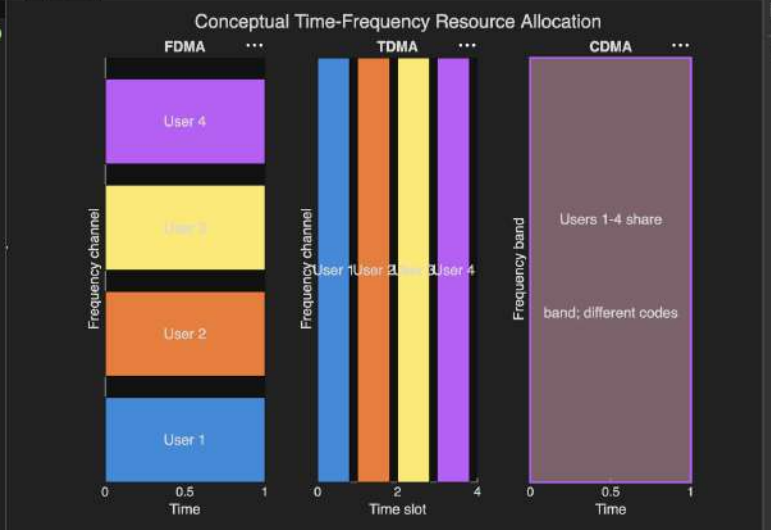
/MATLAB Drive/exp1cg.m

```
figure;
tiledlayout(1,3);
nexttile;
hold on;
colors = lines(4);
for u = 1:4
    rectangle('Position',[0, u-1, 1, 0.8], ...
        'FaceColor',colors(u,:), 'EdgeColor','k');
    text(0.5, u-0.6, sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 1]); ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');

set(gca,'YTick',[]);
hold off;
nexttile;
hold on;
for u = 1:4
    rectangle('Position',[u-1, 0, 0.8, 1], ...
        'FaceColor',colors(u,:), 'EdgeColor','k');
    text(u-0.6, 0.5, sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 4]); ylim([0 1]);
xlabel('Time slot');
ylabel('Frequency channel');
title('TDMA');
set(gca,'YTick',[]);
hold off;
nexttile;
hold on;
for u = 1:4
    rectangle('Position',[0, 0, 1, 1], ...
        'FaceColor',colors(u,:), 'FaceAlpha',0.22, ...
```

Figures

Figure 1



Command Window

```
    generation    system     accessMethod                keyIdea
    __________    _______    ____________    ____________________________________

        "1G"      "AMPS"     "FDMA"          "Separate frequency channels"
        "2G"      "GSM"      "TDMA"          "Separate time slots"
        "2G"      "IS-95"    "CDMA"          "Code-based user separation"
        "3G"      "UMTS"     "WCDMA"         "Wideband code-based access"
        "4G"      "LTE"      "OFDMA"         "Orthogonal frequency resources"
        "5G"      "5G NR"    "OFDMA-based"   "Flexible OFDM-based radio access"

>> exp1cg
>>
```